Architecture Review Board (ARB) Standardization Proposal

EARA-7 Enterprise Agent Reference Architecture

Seven-Layer Technical Architecture & Five Governance Quadrants for Production Readiness

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Five Structural Risks in Production Agent Deployment

5 Blind Spots

Model IQ is rarely the bottleneck; missing perimeter engineering governance drives system failure.

Factual Hallucination & Scope Drift: Guessing internal policies from pre-training memory

Privilege Escalation: Executing tools via super-admin tokens exposes restricted data

Indirect Prompt Injection: Hidden commands in external emails trigger unauthorized writes

Runaway Inference Costs: High-concurrency duplicate queries exhaust flagship token budgets

Regression Black Box: Unverified prompt edits break live safety guardrails overnight

Production Systems Must Align with Level 4 Governance

01

Level 1: Basic Wrapper

Static prompt wrappers without factual verification.

02

Level 2: RAG Assistant

Private knowledge retrieval limited to read-only Q&A.

03

Level 3: Action Agent

MCP tool integration lacking identity and guardrail isolation.

04

Level 4: Governed Agent

EARA-7 zero-trust closed loop; sole production target state.

EARA-7 Seven-Layer Runtime Architecture Topology

L6 Traffic & FinOps Optimization Layer
Semantic caching for frequent queries & dynamic intent routing across SLM and LLM

L5 Bidirectional Guardrails & DLP Layer
Input isolation against prompt injection & output DLP masking for sensitive PII

L4 Identity & Access Governance Layer
End-user OAuth / RBAC token passthrough & zero service-account privilege escalation

L1–L3 Agent Runtime Core Engine
L1 Model & Policy Gateway | L2 Grounded RAG | L3 MCP Read/Write Split & HITL

L7 Observability & LLMOps Layer

End-to-End Trace ID Auditing

Config-as-Code Prompt Versioning

CI/CD Golden Dataset Eval Gates

Five Governance Quadrants & Ownership Matrix

Governance Quadrant	Layers	Owner Team	Key Acceptance KPIs
Q1: Cognitive & Grounding	**L1 – L2**	AI Engineering & Knowledge Ops	Retrieval Recall, Faithfulness Score
Q2: Action & Integration	**L3**	Enterprise Integration & Backend	Tool Success Rate, HITL Interception
Q3: Security & Compliance	**L4 – L5**	InfoSec & Legal Compliance	Zero Escalation, PII Masking Coverage
Q4: Cost & Performance	**L6**	Cloud Infrastructure & FinOps	Cache Hit Rate, Cost per Session
Q5: Reliability & LLMOps	**L7**	Platform Engineering & SRE / QA	Trace Coverage, CI/CD Eval Pass Rate

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Four Defense-in-Depth Controls for Enterprise Audits

L4 Zero-Trust Identity

Pass end-user tokens to enforce granular RBAC checks.

L5 Untrusted Input Isolation

Separate system policies from untrusted external payloads.

L5 Output DLP Redaction

Automatically mask PII and financial credentials on egress.

L3 High-Risk HITL Gate

Pause fund transfers and core writes for human approval.

Shared Platform Core with Pluggable Domain Packs

Finance & Compliance
Regulatory RAG & audit pre-check tools

Supply Chain Ops
Equipment manuals & ERP reorder workflows

Customer Support
Policy knowledge & order lookup tools

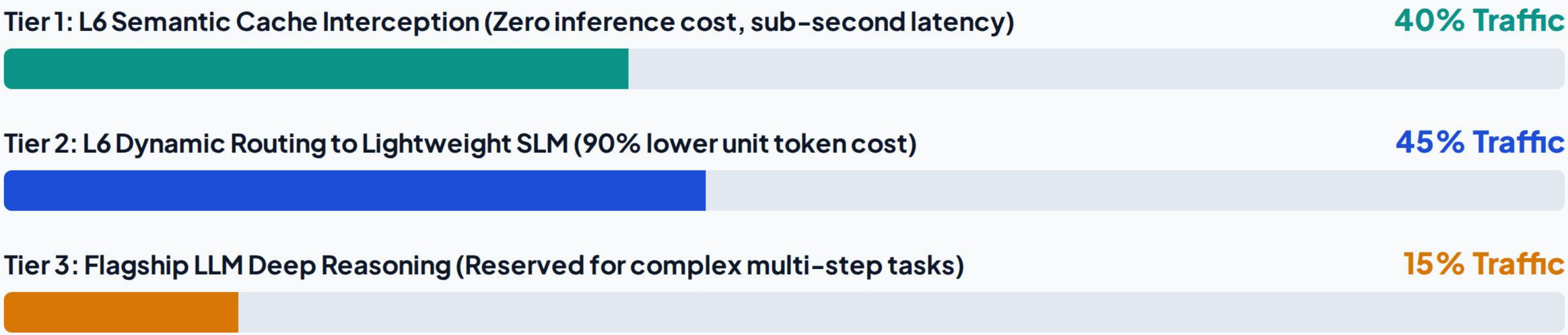
HR & Internal IT
Employee handbooks & access provisioning

▲ Business units configure L2 knowledge indexes and L3 MCP tools to launch in days ▲

Enterprise Shared Platform Core (L1 Gateway · L4 Identity · L5 Guardrails · L6 FinOps · L7 LLMOps)

Centrally operated by Platform Engineering so every domain agent inherits baseline security and observability

Two-Stage Traffic Funnel Cuts Inference Spend by 60%–75%



Paired with tenant-level token budget caps and fallback circuit breakers to prevent peak-traffic bill spikes.

Three Value Levers Achieve 6–9 Month Payback

6–9 Mos
Projected Platform Payback Period

70%
Redundant Core Engineering Saved

65%
Net Annual Inference Cost Reduction

Platform Reuse Savings

(N – 1) × Standalone Core Cost × 70%

FinOps Inference Savings

Annual Requests × (LLM Price – Blended Price)

Efficiency & Risk Avoidance

Automated ticket hours saved & zero breaches

12-Week Three-Phase Rollout & ARB Approval Request

Phase 1 (Weeks 1–4)

Deploy model gateway, grounded RAG, and global Trace IDs

Phase 2 (Weeks 5–8)

Enable MCP tools, RBAC passthrough, and security guardrails

Phase 3 (Weeks 9–12)

Launch semantic caching, routing, and CI/CD eval gates

Formal ARB Decision Request (Ask for Approval)

1. **Standard Ratification:** Adopt EARA-7 & Five Governance Quadrants as mandatory enterprise standard
2. **Production Gate:** Authorize Platform & InfoSec teams to enforce pre-launch security and eval audits
3. **Resource Allocation:** Approve 12-week core platform build budget and 1-2 pilot domain workstreams